Integrating psycholinguistic evidence into Japanese readability assessment

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Research on Japanese readability has largely relied on surface-level metrics such as sentence length or character type ratios (Lee, 2016; Shibasaki & Hara, 2010), leaving open whether these measures capture the cognitive processes involved in reading. Recent English study shows that psycholinguistically grounded indicators, especially surprisal, predict eye-movement behavior more strongly than traditional readability formulas (Klein et al., 2025). It is unknown whether these findings extend to Japanese, whose mixed writing system (*hiragana*, *katakana* and *kanji*) introduces distinct orthographic processing demands (Nakayama 2017). This gap raises the question of whether surprisal-based predictors offer a more cognitively valid account of Japanese readability.

To test this prediction, we reanalyzed BCCWJ-EyeTrack (Asahara et al., 2019), which includes eye-tracking and self-paced reading data from 24 native speakers reading 20 news articles (218 sentences). Articles were drawn from the newspaper domain of BCCWJ, covering politics, economy, and general news. We compared traditional Japanese readability formulas (ARI, Lee-formula, Shibasaki-formula) with psycholinguistic indicators such as lexical density, word frequency, word length and surprisal. These variables were selected due to their established roles in orthographic processing, lexical access, and information density in psycholinguistics. Surprisal was computed from a transformer-based language model and normalized by base-phrase *bunsetsu* units and tokens to account for different Japanese-specific segmentation. Reading time measures (e.g., First Pass Time, First Fixation Time, Total Time) were aligned to the same units. *Bunsetsu* units were used because they constitute the basic annotated unit in BCCWJ-EyeTrack and correspond to prosodic and syntactic processing units in Japanese reading.

Across sentence- and paragraph-level analyses, *bunsetsu*-normalized surprisal showed the strongest and statistically reliable correlation with reading time measures, replicating the surprisal advantage reported for English (Klein et al., 2025). Among traditional metrics, Shibasaki's predicate-based formula showed the highest alignment with cognitive load, while other formulas exhibited weak associations. Word frequency correlated negatively with reading times, confirming its role in facilitating lexical access.

These results demonstrate that average per-*bunsetsu* surprisal, a psycholinguistically grounded feature, correlates with Japanese eye-movement data to a degree comparable to mean surprisal in English. This parallel, supported by the role of *bunsetsu* as the basic processing unit in BCCWJ-EyeTrack, suggests that surprisal-based predictors tap into processing mechanisms that generalize across typologically and orthographically distinct languages, offering an operational framework for cross-linguistic readability assessment. The findings highlight the need for reader-specific behavioral data from larger and more diverse reader populations and motivate the development of psycholinguistically informed Japanese readability corpora.

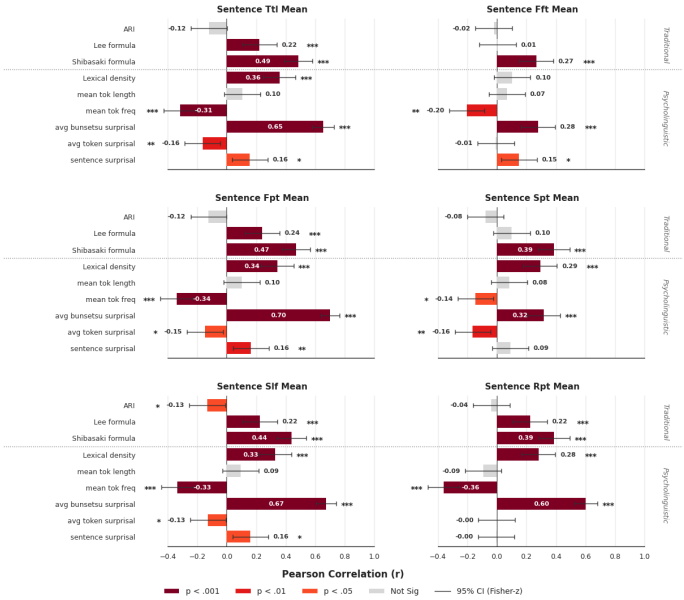

Figure1. Sentence-level Pearson's correlations between reading times and features. Error bars indicate 95% confidence intervals based on Fisher's z transformation.

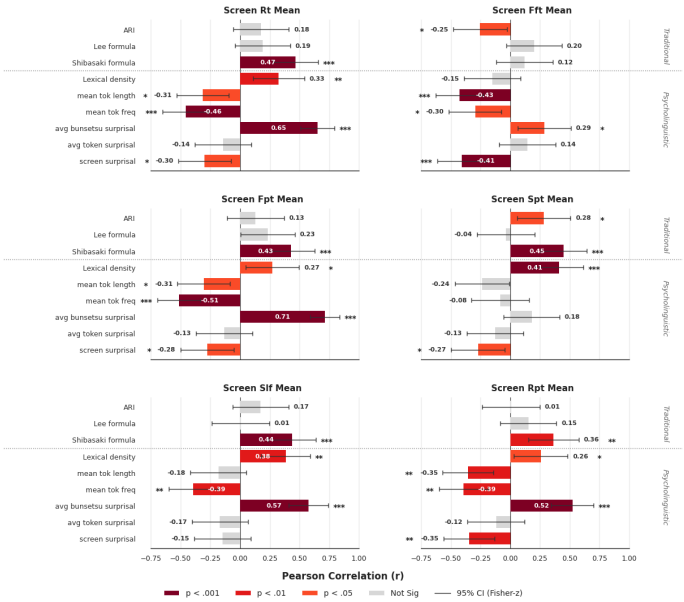

Figure2. Paragraph-level Pearson's correlations between reading times and features. Error bars indicate 95% confidence intervals based on Fisher's z transformation.

References

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